Machine Learning

Subject: Computer Science | NoteShare Library

MACHINE LEARNING

1. TYPES
Supervised, Unsupervised, Reinforcement, Semi

2. SUPERVISED
Linear Regression:

Gradient descent

Logistic: sigmoid(z) = 1/(1+e^-z)
Decision Trees: Gini, Entropy
Random Forest: ensemble
SVM: max-margin, kernels
KNN: lazy, majority
Naive Bayes: independence

3. UNSUPERVISED
K-Means: centroid iteration
Hierarchical: dendrogram
DBSCAN: density
PCA: eigenvectors

4. NEURAL NETWORKS
Perceptron
Activations: Sigmoid, Tanh, ReLU, Softmax
Backprop
Optimizers: SGD, Adam
Regularization: L1, L2, Dropout

5. DEEP LEARNING
CNN: images
RNN: sequences
LSTM: long deps
Transformer: attention

6. EVALUATION
Accuracy, Precision, Recall, F1
MSE, RMSE, MAE, R
Cross-validation

7. FEATURES
Normalization, Standardization
One-hot, Polynomial
SMOTE